Julius Bär

BREAK THE SILO

How we built a decentralised control plane for service orchestration

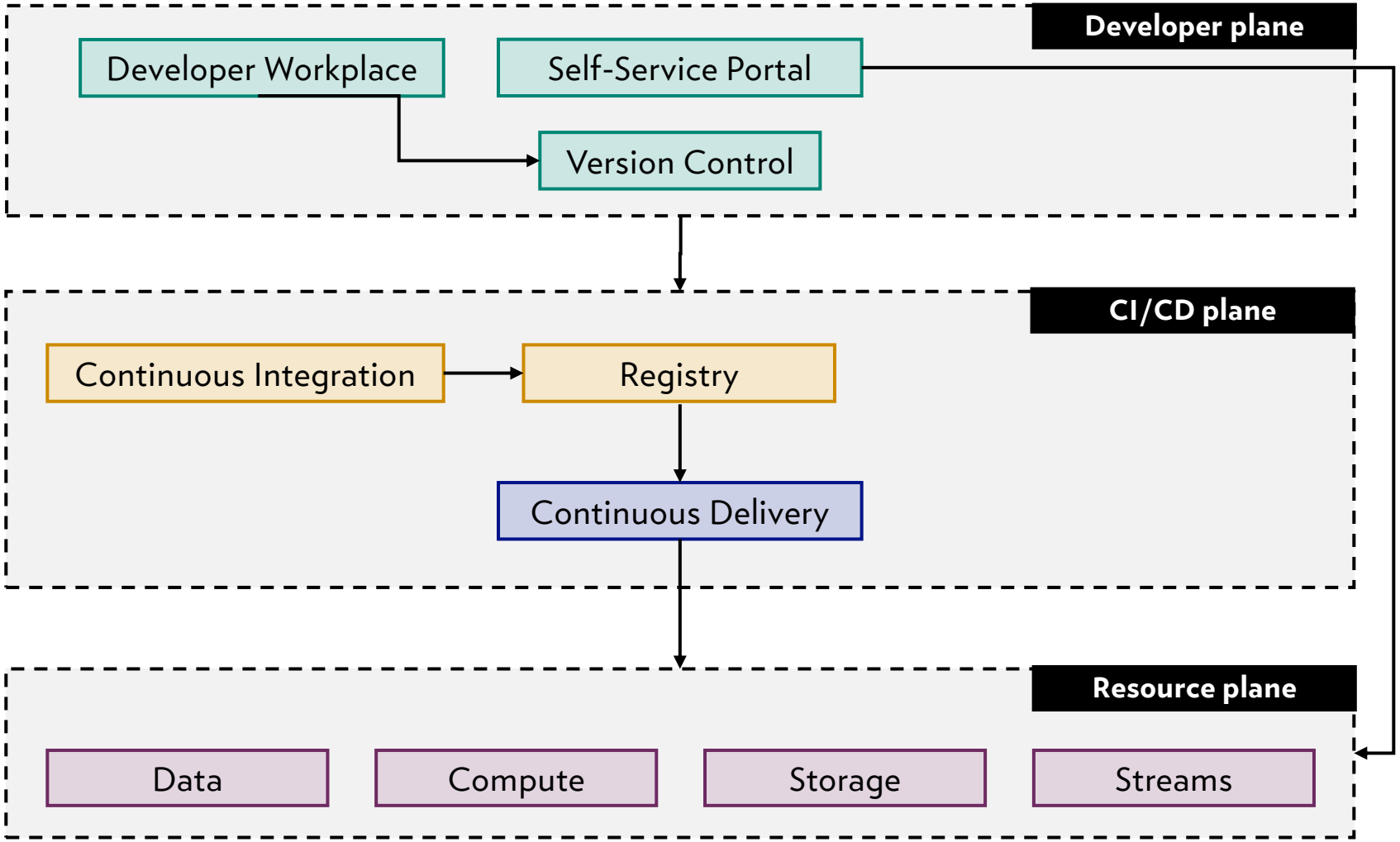
Voxxed Days Zurich 2026

By Eros Taborelli – Principal Engineer & Software Architect at Julius Baer

WHY?

Context

Internal Developer Platform



Focused on Development

- 800+ engineers
- hundreds of applications
- Self-Service

Where we come from

The ‘traditional’ way of ordering infrastructure services

An engineer

Makes an order on a portal

And receives a resource
(example: VM)

Why it works

Benefits of a UI

- Filling a form is a normal, mundane task
- Immediate validation and visual feedback
- Some products and services come with a portal by default

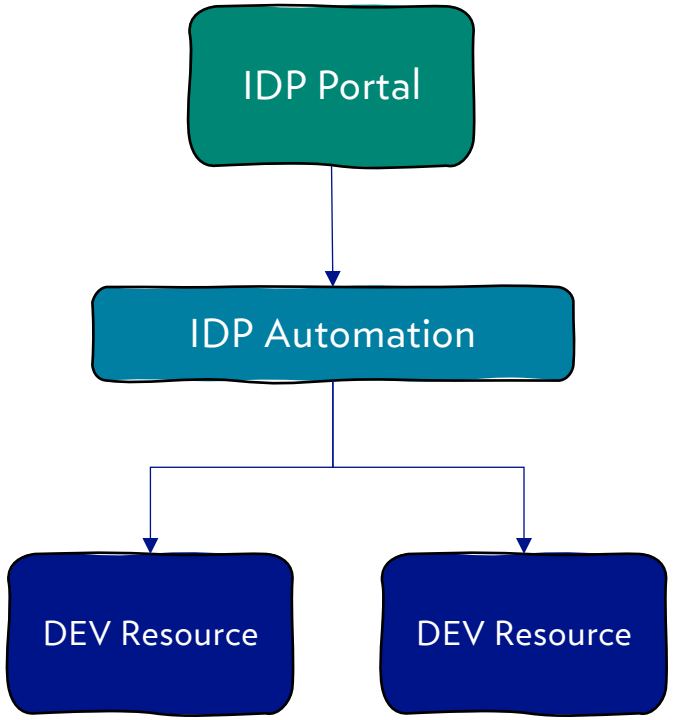
Shortcomings

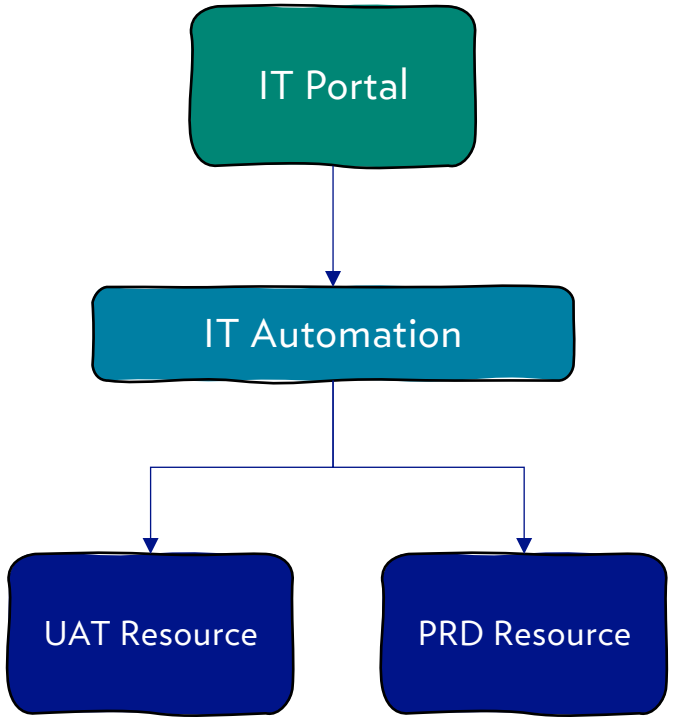
Hidden costs of a UI

- Day 2 operations must be implemented
- No comparison between environments
- Repeatability/Replicability

Where we come from

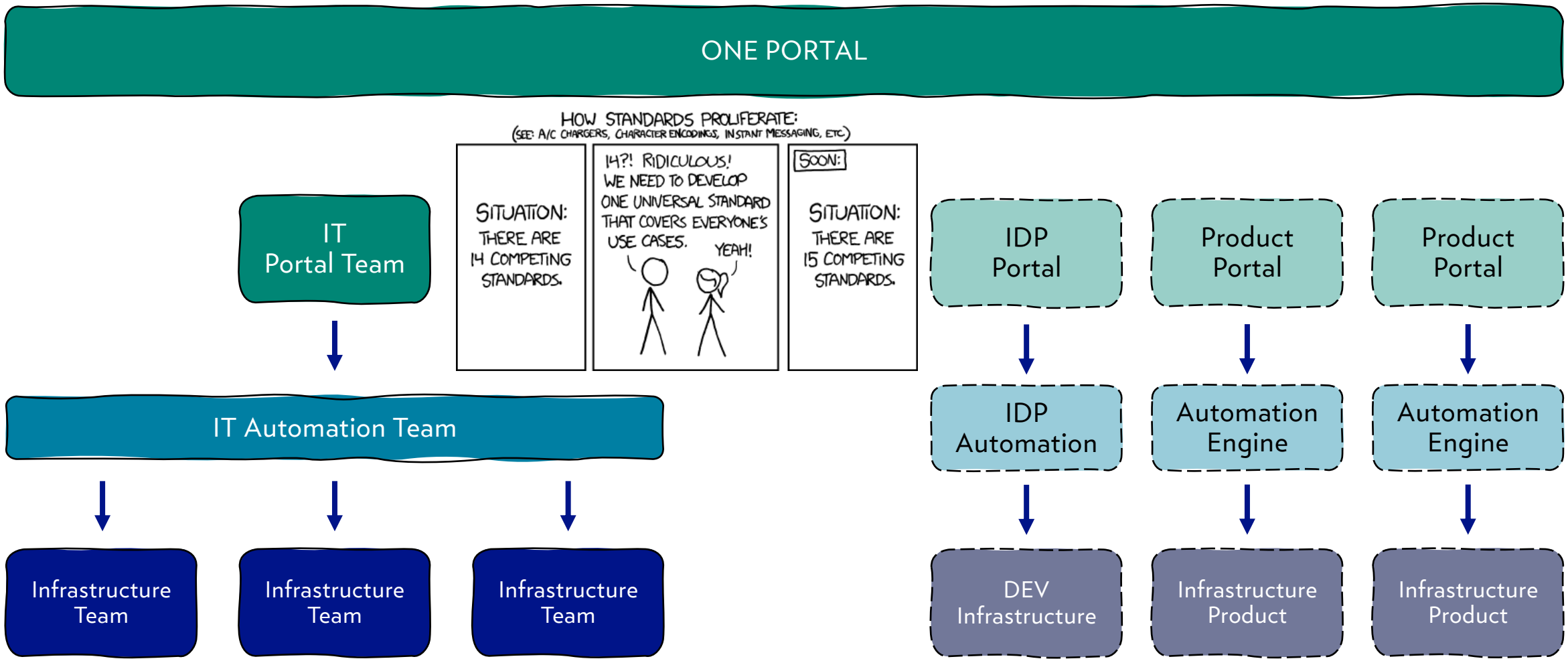
What about higher (test, staging, uat, prod) environments?





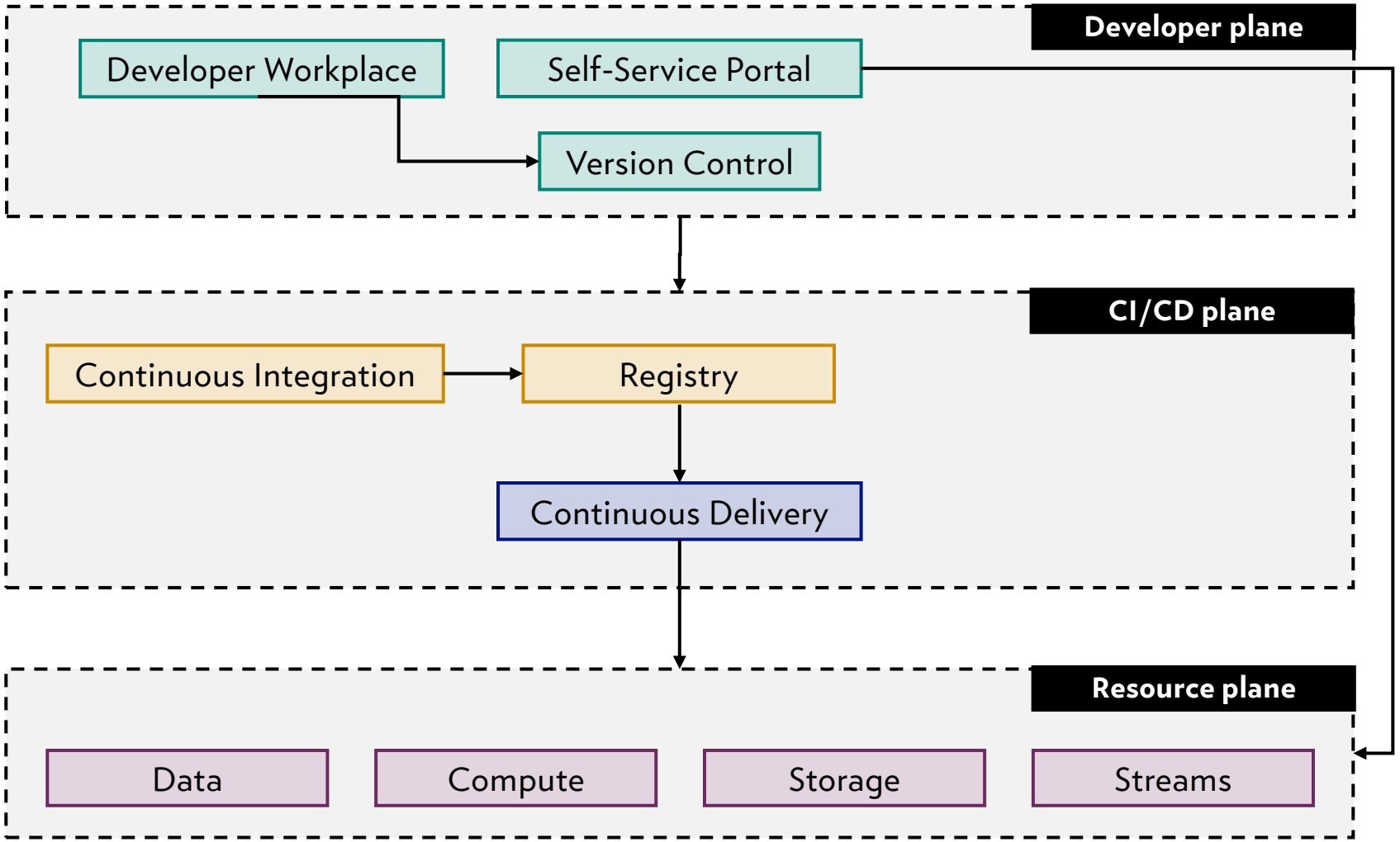
Split Responsibilities

Multiple teams involved, unclear ownership for the full process



Context

Internal Developer Platform



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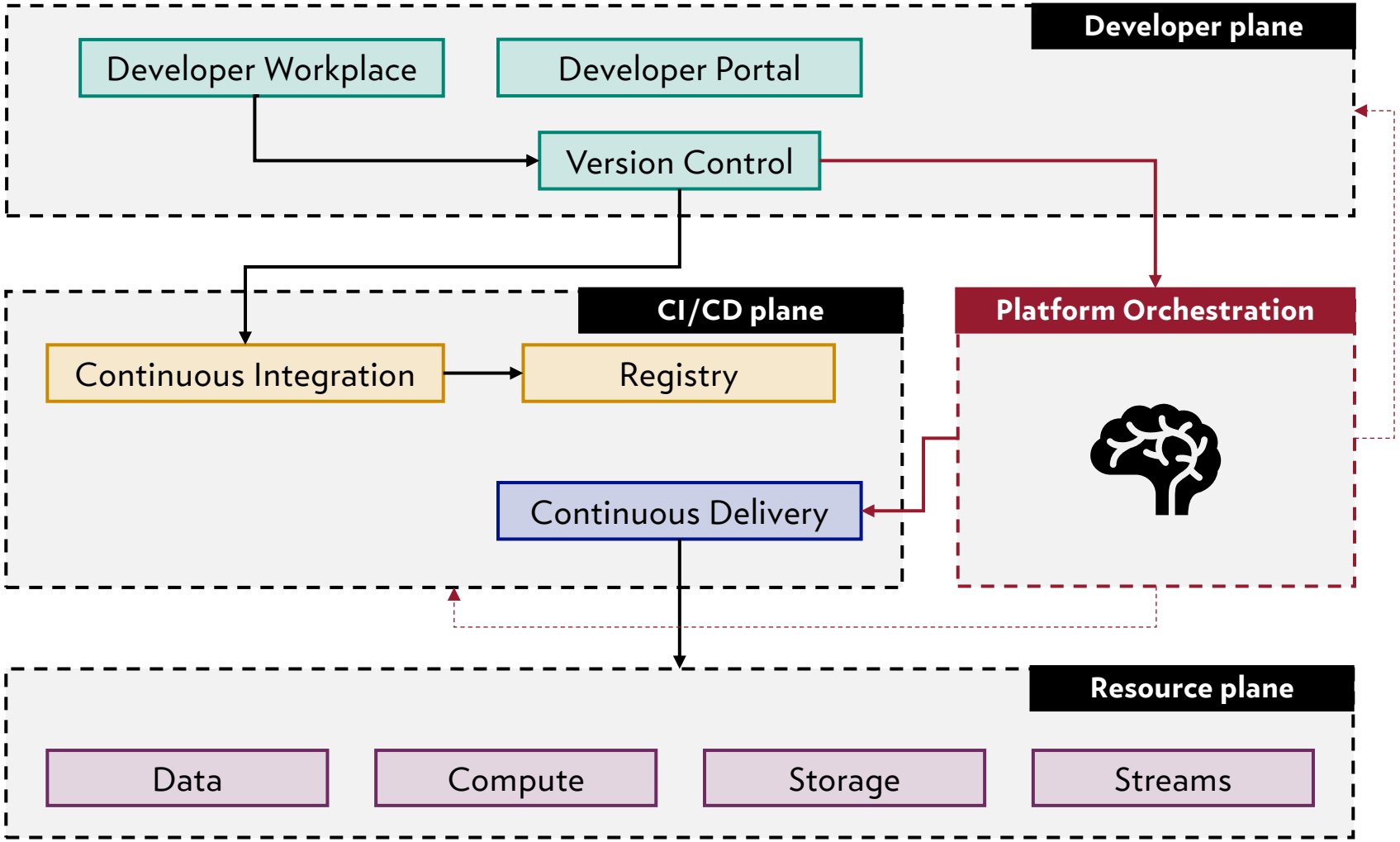
Context

Internal DevOps Platform



Streamlined Experience

- from DEV to PRD
- Self-Service remains

Our goals

Standardisation

Consistent experience when consuming infrastructure

Operational excellence

Actively promote the usage of Infra-as-Code

Ownership

Infrastructure Teams to drive the offering to end customers

WHY?

Our goals

Standardisation

Consistent experience when consuming infrastructure

- Familiar APIs with common patterns: Service Catalog
- No key differences when working with a database, a Kafka topic, or an object storage

Operational excellence

Actively promote the usage of Infra-as-Code

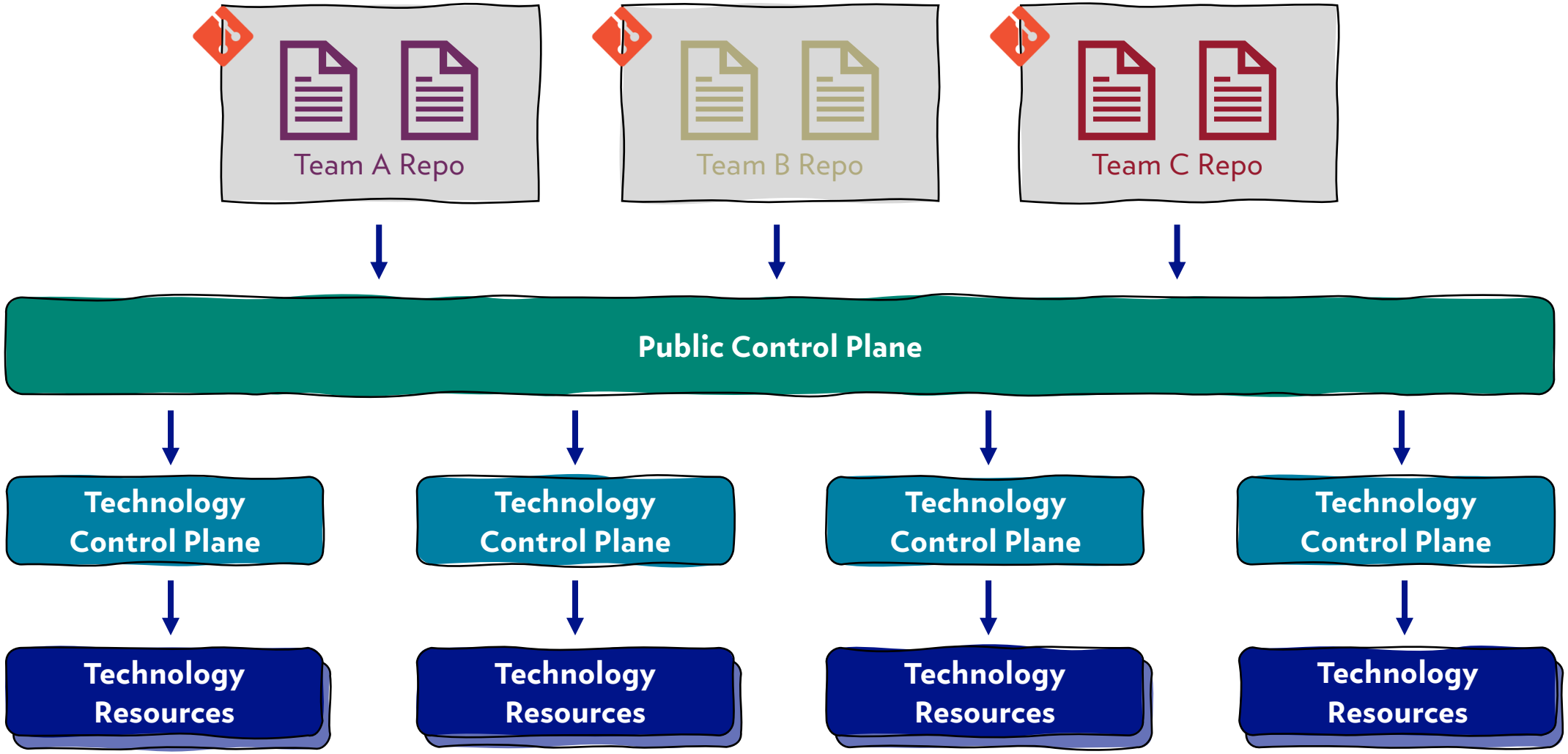
- Goodbye ClickOps
- Repeatable Infrastructure
- Built-In audit, diff, and day2 operations
- Apply Software Development best practices to infrastructure

Ownership

Infrastructure Teams to drive the offering to end customers

- Reduce dependencies
- End-to-End experience owned by the infrastructure teams, regardless of organizational structures
- More direct customer feedback

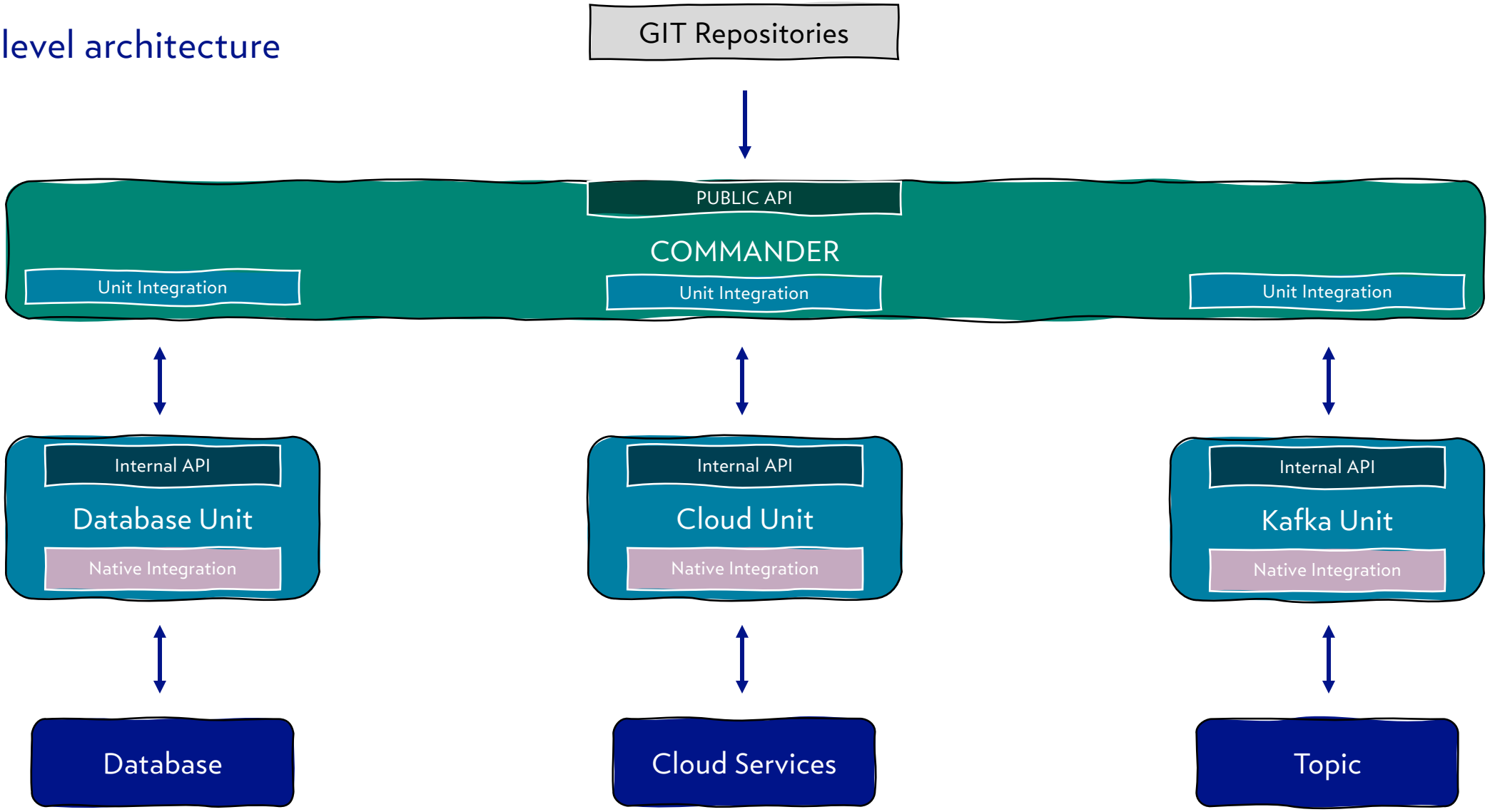
Concept


Team A Repo
Team B Repo
Team C Repo
Public Control Plane
Technology Control Plane
Technology Control Plane
Technology Control Plane
Technology Control Plane
Technology Resources
Technology Resources
Technology Resources
Technology Resources

WELCOME: FLEET COMMANDER

Fleet commander

High level architecture



Goals achieved?

- ✓ Operational Excellence: GitOps / X-as-Code
- ✓ Standardization: Service Catalog
- ? Ownership: Decentralized Architecture and Responsibilities

KUBERNETES

+

CROSSPLANE

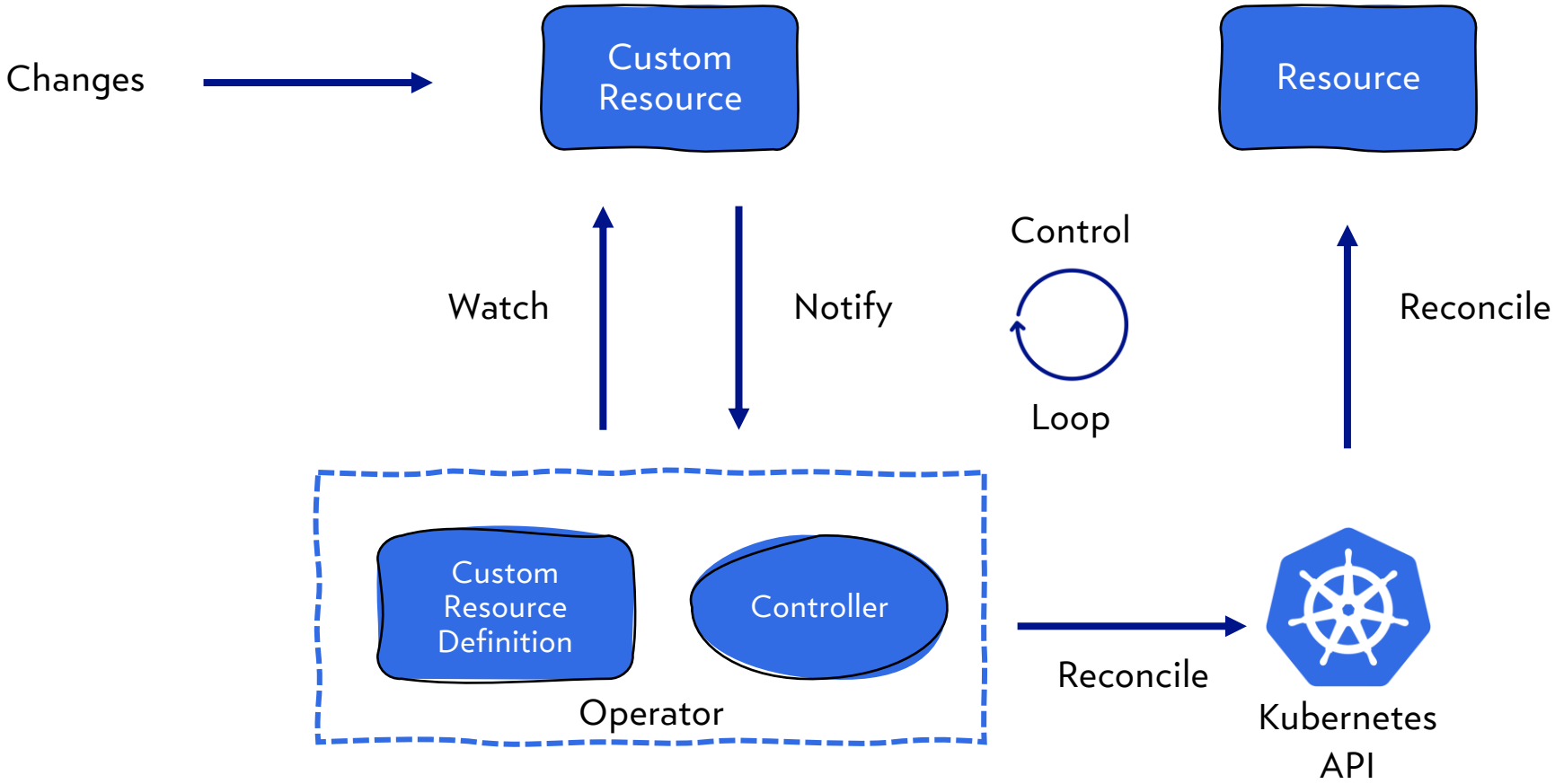
=

FLEET COMMANDER

WHY?

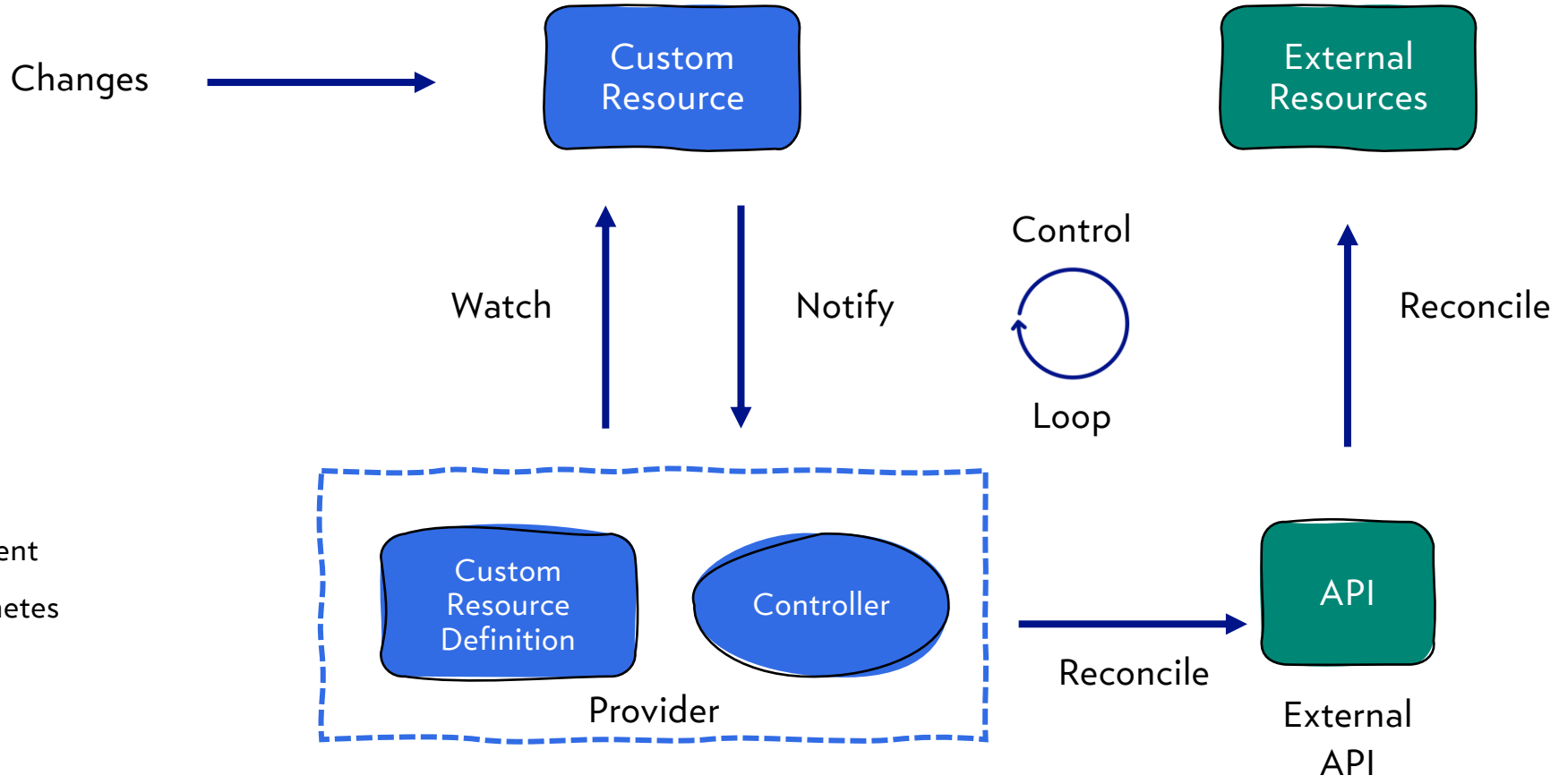
Operator pattern

Control loop on custom resources



Crossplane providers

Operators for external resources



Examples

- provider-azure
- provider-aws
- provider-gcp
- provider-confluent
- provider-kubernetes
- provider-vault

Crossplane composition

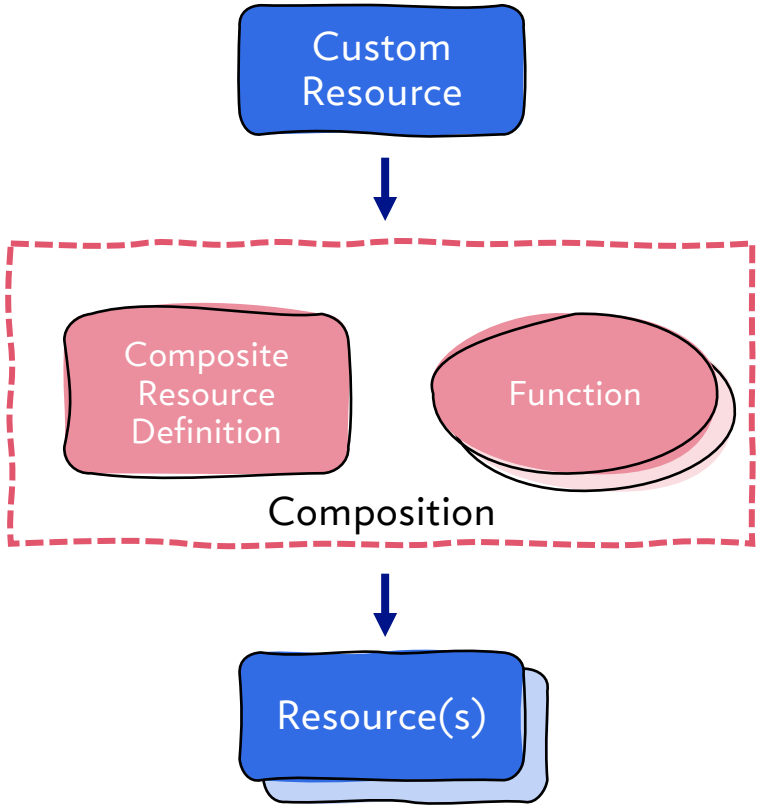
Harness the power of custom resource definitions

Composite resource definition

- Define the **Public API** of the Composition.
- When the Composition is deployed, Crossplane creates a standard Kubernetes **Custom Resource Definition**.

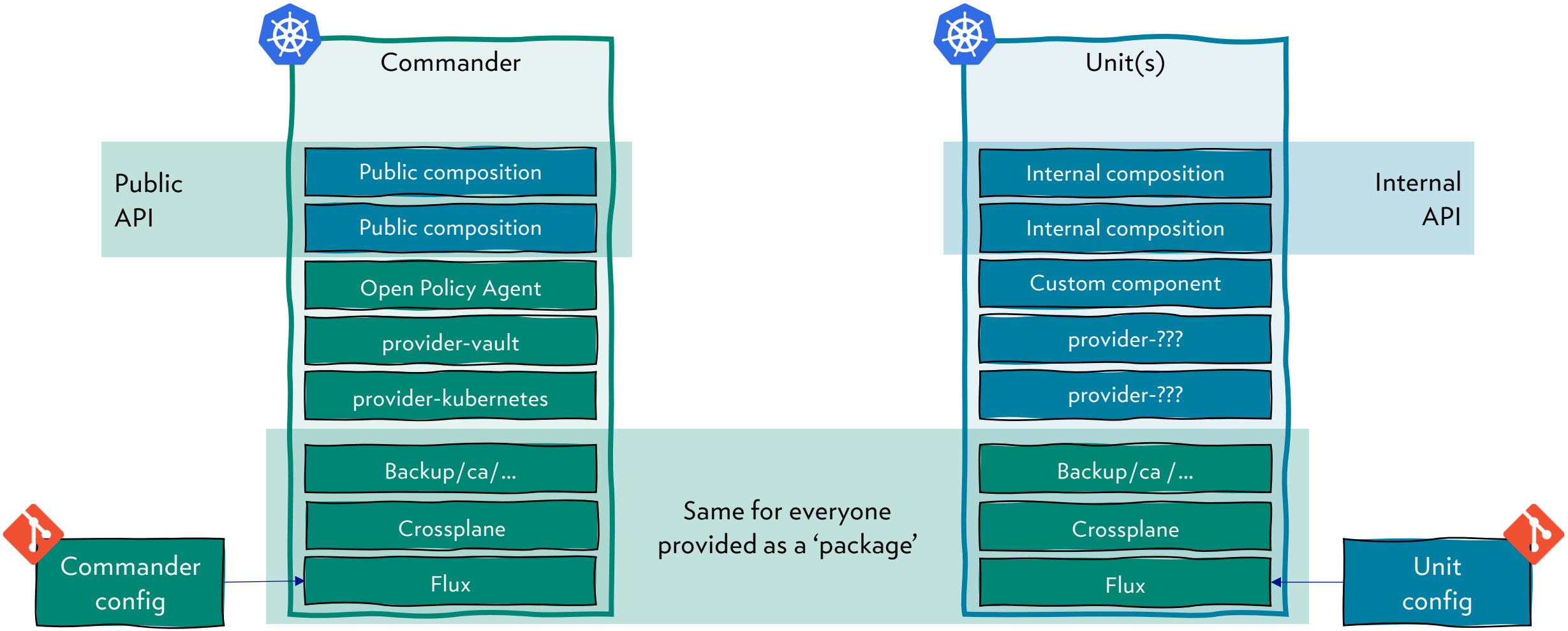
Function

- The implementation of the composition.
- Handles the CR and **decompose to one or multiple Kubernetes resources.**
- (hint: including resources defined by Crossplane providers and other compositions)



Fleet commander

Shared responsibility model

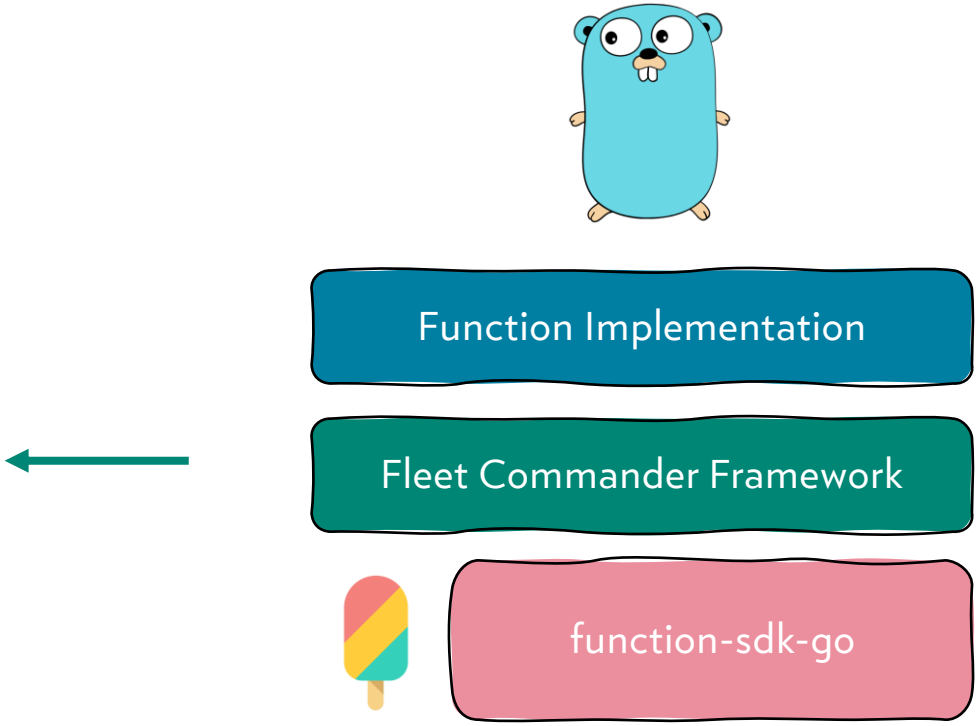


SDK

What fleet commander offers for easier onboarding

Fleet commander framework

- Golang: type safety, XRD generation, sustainable long-term
- Utility functions
  - provider-kubernetes wrapper (more later)
  - labels & annotations
  - unit testing
- Generic Helm chart (unit configuration)
- Build system



Goals achieved?

- ✓ Operational Excellence: GitOps / X-as-Code
- ✓ Standardization: Service Catalog
- ✓ Ownership: Decentralized Architecture and Responsibilities

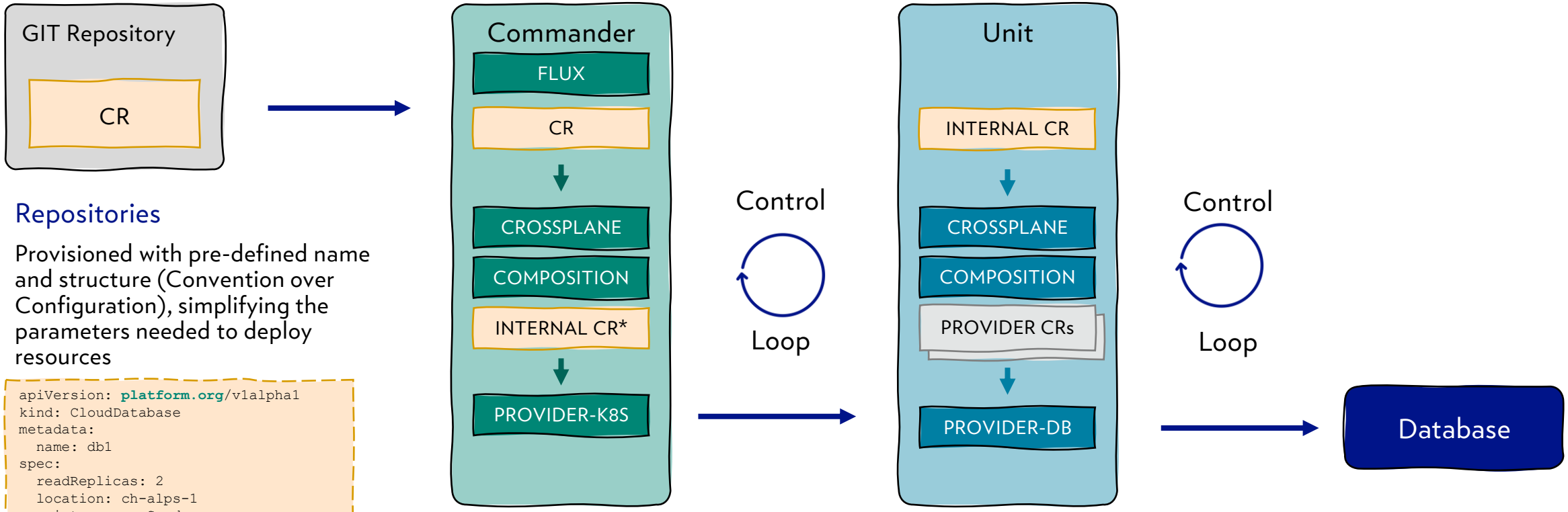
BUT HOW DOES IT WORK?

Example

Provision a generic resource

*: wrapped for the k8s provider

```yaml
apiVersion: kubernetes.crossplane.io/v1alpha2
kind: Object
spec:
  forProvider:
    apiVersion: internal.platform.org/v1alpha1
    kind: CloudDatabase
    spec:
      readReplicas: 2
      location: ch-alps-1
      maintenance: Sunday
  providerConfigRef:
    name: dbunit
```



Repositories

Provisioned with pre-defined name and structure (Convention over Configuration), simplifying the parameters needed to deploy resources

```yaml
apiVersion: platform.org/v1alpha1
kind: CloudDatabase
metadata:
  name: db1
spec:
  readReplicas: 2
  location: ch-alps-1
  maintenance: Sunday
```

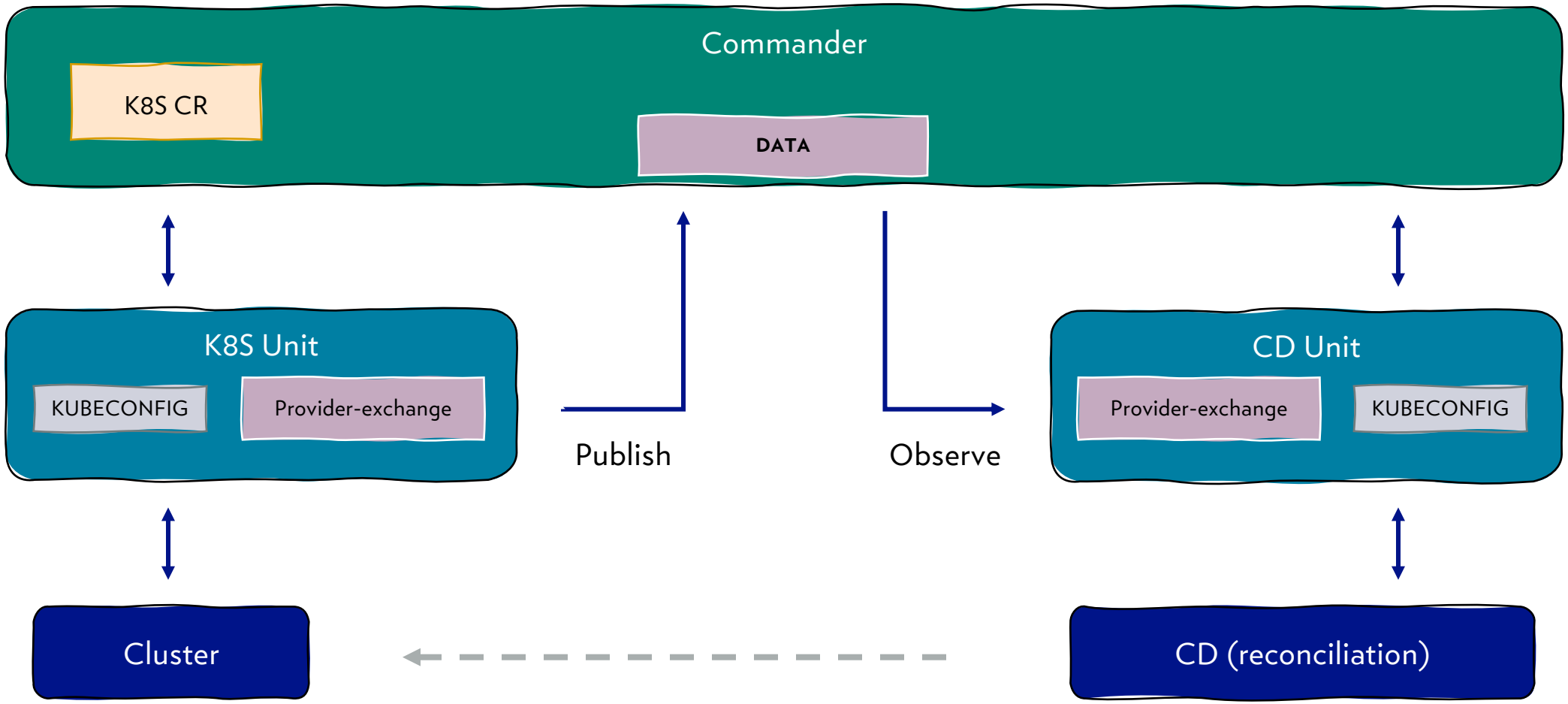
DEMO TIME

github.com/erost/vdz26-demo-fleet-commander

CHALLENGES

Data exchange

Solve dependencies between units (example)



Lifecycle

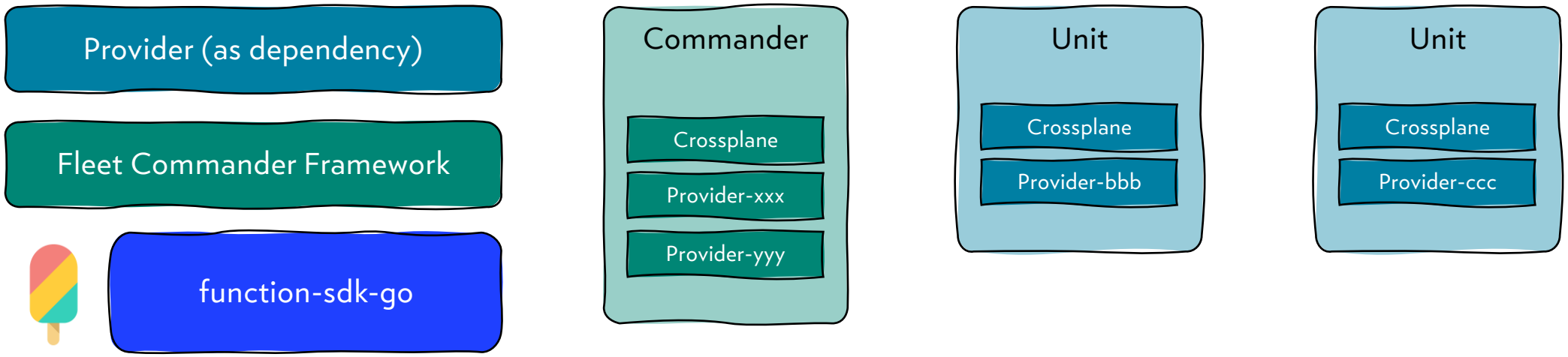
How to handle Crossplane updates in a distributed environment

Crossplane versions

- Crossplane runtime compatibility
- Crossplane libraries compatibility
- Fleet Commander Framework compatibility

Providers' requirements

- Providers have different Crossplane version requirements
- Different providers may have conflicting requirements



Onboarding costs

Steep learning curve

Learn Kubernetes concepts

- CRDs
- Operators

Learn Crossplane

- Providers
- Compositions
- Functions

Get familiar with Fleet Commander

- Architecture
- Framework
- Golang

Operate Unit

- Kubernetes
- Dependencies
- Maintenance

Q&A

THANK YOU!

github.com/juliusbaer/conferences

eros.taborelli@juliusbaer.com

Fleet Commander building blocks

Crossplane

Main project and releases

https://github.com/crossplane/crossplane

Develop compositions using Go

https://github.com/crossplane/function-sdk-go

Kubernetes Provider

Deploy and Manage arbitrary Kubernetes objects on different clusters (commander to unit)https://github.com/crossplane-contrib/provider-kubernetes

two Kubernetes

commander

- crossplane
- provider-kubernetes
  - configure unit credentials

- write public composition
  - decomposes into a kubernetes object
  - kubernetes object wraps internal composition custom resource
  - reference the unit configuration to delegate

- apply custom resource

unit

- crossplane
- desired providers

- write internal composition
  - decomposes into provider custom resources

- it appears here